EXPERIMENT NO: 8
User Management and Role-Based Access Control

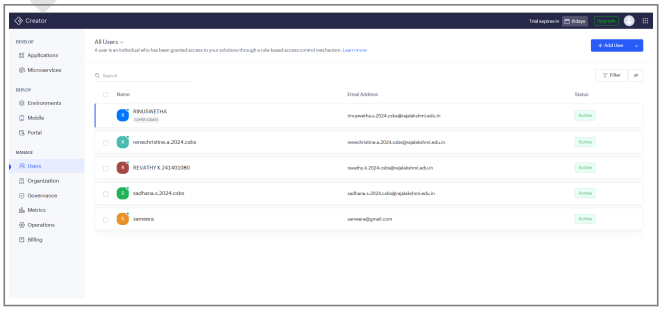
AIM

To implement user management and role-based access control (RBAC) in the Event Registration application using Zoho Creator, by adding users, defining permission profiles, creating a role hierarchy, and configuring data sharing rules.

PROCEDURE

1. Open Zoho Creator and go to **Manage > Users** from the left navigation panel. The list of existing users is displayed. Only the Super Admin or an authorised administrator can open this section.
2. Click **Add User**. In the panel that opens, enter the **Email Address** and **Name** of the new user, and tick *Send Notification Mail* if an invitation is required.
3. Under **Add to Application** (tab *Applications as User*), select the **Application Name** (Event Registration), choose the **Role** and choose the **Permission** for the user. Click **Add User** to save. The new user receives access to the application with exactly the rights of the chosen role and permission.
4. Open the application and go to **Settings > User Permissions > Permissions**. Here the permission profiles *Read* and *Write* are available; new profiles can be added with **Add Permission**. Every user in the application must be linked to one of these permission profiles.
5. Go to the **Roles** tab and build the role hierarchy (CEO > manager > employees) using **Add Role** and **Manage Role Hierarchy**. The hierarchy decides which users can see the records of other users.
6. Go to the **Data Sharing** tab, click **New Share Rule**, and specify the form, the user/role the data is shared from, the role it is shared to, and the level of access (Read/Write). Save the rule; it becomes active immediately for the selected role.

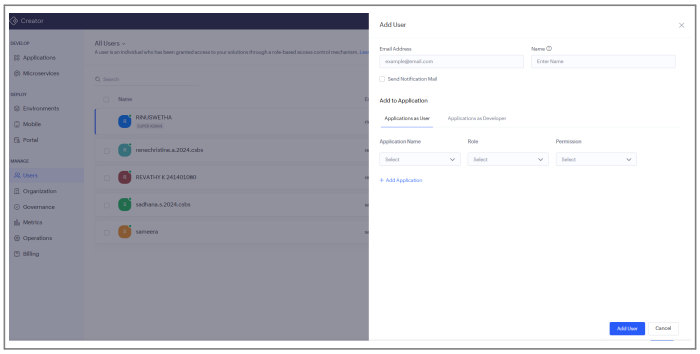
Step 1 – All Users list



The **All Users** page under **Manage > Users** lists every user who has been given access to the solutions through role-based access control. It shows each user's name, email address and status. **RINUSWETHA** is the **Super Admin**, and the other users (renechristine, REVATHY K, sadhana and sameera) are all **Active**.

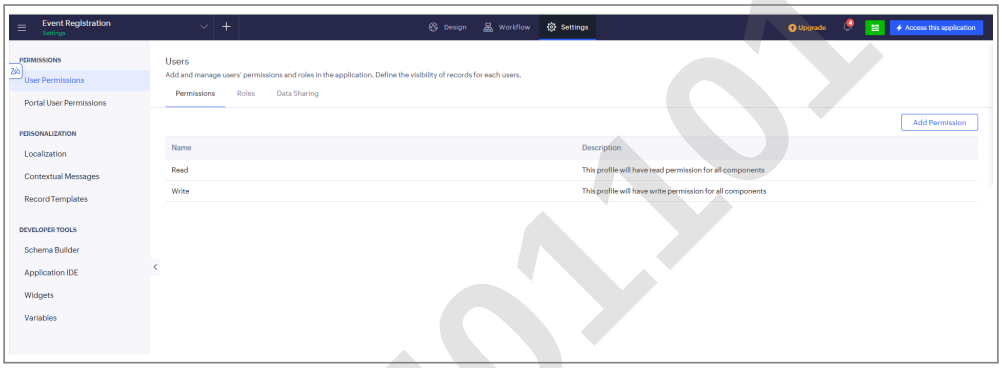
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Step 2 & 3 – Adding a user to the application



The **Add User** panel in Zoho Creator collects the user's email address and name, and lets the administrator assign the application, the role and the permission under *Applications as User*. The left panel shows existing users such as RINUSWETHA (Super Admin) and others.

Step 4 – Permission profiles



Under **Settings > User Permissions > Permissions**, the application has two profiles: **Read** (read permission for all components) and **Write** (write permission for all components). The **Add Permission** button creates custom profiles.

OBSERVATIONS ON THE PROCEDURE

While adding a user, the email address uniquely identifies the person, and the notification mail informs them that they have been invited to the application. Without selecting an application, role and permission, the user exists in the account but cannot open the Event Registration application. Once the role and permission are chosen, the user can log in and works only within the limits set by the administrator.

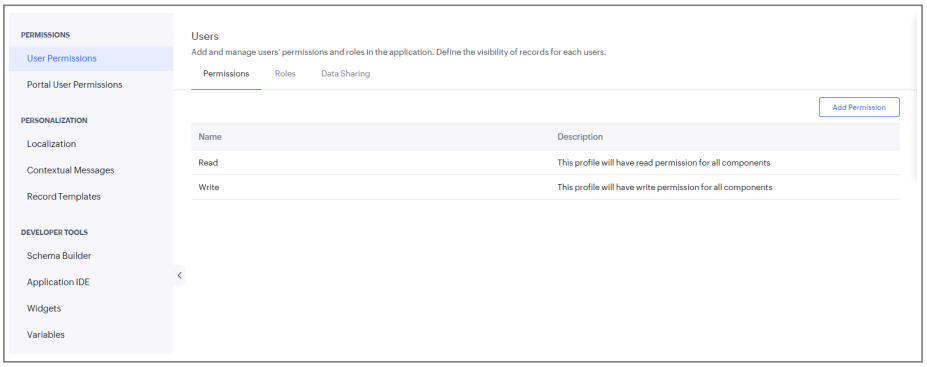
The All Users page gives a quick view of everyone who can access the account. The Super Admin can search for a user, filter the list, and check whether each user is Active before assigning a role or permission. The Read and Write permission profiles separate people who only need to view registrations from people who must create or edit them, and custom profiles can be created with Add Permission when finer control over individual forms, reports or pages is needed.

Permissions, roles and data sharing rules are configured separately, so a change in one of them does not disturb the others. This makes access control flexible: a user can be moved to a different role without redefining the permission profile, and a new share rule can be added without changing the role hierarchy.

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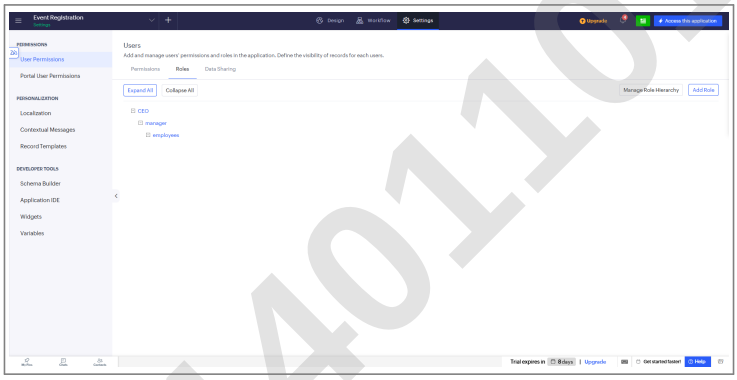
OUTPUT

Output 1 – Permissions



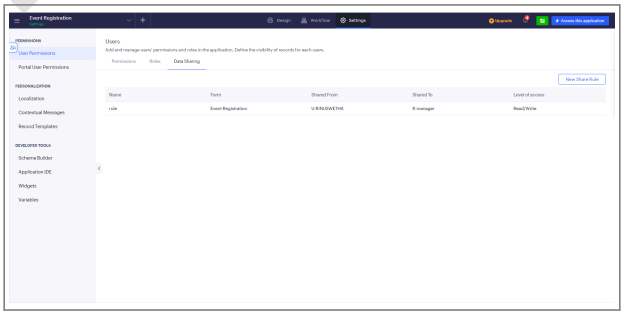
The **Permissions** tab lists the available permission profiles. *Read* grants read access and *Write* grants write access to all components of the application. These profiles are assigned to users while adding them.

Output 2 – Roles (Role Hierarchy)



The **Roles** tab shows the role hierarchy of the application: **CEO** at the top, **manager** below the CEO, and **employees** at the lowest level. Users higher in the hierarchy can view the records of the roles beneath them. Roles can be added with *Add Role* and rearranged with *Manage Role Hierarchy*.

Output 3 – Data Sharing



The **Data Sharing** tab shows the share rule named **rule** created for the form **Event Registration**. Records of the user **RINUSWETHA** (Shared From) are shared with the role **manager** (Shared To) with **Read/Write** access. This defines the visibility of records for each user beyond the default role hierarchy.

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RESULT

Thus, user management and role-based access control were successfully implemented in the Event Registration application. Users were added with assigned roles and permissions, a role hierarchy (CEO, manager, employees) was created, and a data sharing rule was configured to control record visibility and access levels.

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